

Prepare a Pet Feeding Station SOP

One episode prepares one feeding station. It runs from a debris covered mat, two empty bowls, an open kibble bag and a filled pitcher, to a wiped mat carrying a half full water bowl on its left half and a half full food bowl on its right half, with the kibble bag folded and clipped shut and every tool back on its spot.

The order never changes: **wipe the mat, stage the two empty bowls on it, scoop the kibble, seal the bag, pour the water.** Nothing goes onto the mat until the mat is clean, and both bowls are filled where they stand on the mat, so no full bowl is ever carried.

The right gripper does the primary work: the wipe, the scoop and the clip, and it carries the food bowl. The left gripper supports: it holds the bag while the scoop works in it, holds the fold while the clip goes on, and it owns the water side, carrying the water bowl and the pitcher. The one two armed action is the seal: both grippers straighten the bag's top edge and turn the fold down together, each on its own end, before the right gripper leaves to fetch the clip. The right arm is the working arm because the rig is set up right handed, so the dominant hand drives it in leader follower. A left handed collection requires a separately approved mirrored variant. Do not swap arms during an episode.

Neither gripper reaches onto the other arm's side to pick anything up or to carry it. Two objects sit on the midline and are shared: the kibble bag, which each gripper meets from its own side, the right from the right and the left from the left, and the feeding mat, which the right gripper wipes across its full width. Nothing is ever passed from one gripper to the other, only one object is held at a time, and no grasp is changed while an object is off the surface.

Each tool goes back to its own spot before the next step starts. The counts are fixed: two level scoops of kibble and one continuous water pour, each bowl filled to roughly half. There is no correction scoop and no top up pour. Kibble or water that lands outside a bowl is left exactly where it falls, tagged, and the episode continues. Nothing is ever swept up on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.

2. The environment camera frame holds the whole tabletop in one shot: the pitcher stand, the bag spot, the scoop spot, both bowl spots, the mat, the crumb spot and the cloth spot.
3. The pitcher spout and the water bowl rim both stay in frame for the whole pour, and the debris on the mat is visible before the wipe.
4. Both arms are at home with their grippers open.
5. The tabletop is dry and clear of anything but the items in the Materials checklist.
6. The right arm reaches the scoop spot, the bag from its right side, the food bowl spot, the cloth spot and the full width of the mat without extending to a joint limit or folding in on itself. Narrow the mat or bring it forward until this is true.
7. The left arm reaches the pitcher stand, the bag from its left side, the water bowl spot and the left half of the mat without extending to a joint limit.
8. Neither arm needs the far side of the table for anything. The paths from the scoop spot to the bag to the mat, from the pitcher stand to the mat, and from each bowl spot to the mat are unobstructed. Carry each empty bowl forward by hand during setup: it must clear the standing bag on both sides. Widen the gap between the bag and the bowl spots until it does.

Materials checklist

1. One flat mat lies at the front center of the table, straddling the midline, bare of bowls, and no wider than the two bowls standing side by side with a gap between them.
2. The mat grips the table. Draw a cloth across it by hand during setup: if it slides, bunches or lifts, weight it or replace it with a rubber mat that stays put. Nothing pins the mat during the episode.
3. Debris is scattered across the mat, loose kibble and dry crumbs, spread out rather than clumped, none of it touching the mat edges.
4. The crumb spot in front of the mat is bare.
5. One dry cloth, folded flat, lies on the cloth spot.
6. Two bowls stand empty and dry, the water bowl on its spot left of the midline and the food bowl on its spot right of the midline. The two are clearly different in size or colour, and the same bowl is the food bowl in every episode. Each bowl has a rim the gripper can close on squarely, and is low enough and wide enough to hang upright and level from that grip.
7. The kibble bag stands open on its own at the bag spot, between the two bowl spots and clear of both, low enough that the scoop reaches the kibble without the gripper passing below the bag's top edge, and holding well more than two scoops. It is no taller than it is stable: a bag a passing bowl could brush over is replaced or brought forward.
8. The bag's top edge is soft and holds a crease. Crease it once by hand during the reset so the fold lands in the same place every episode.
9. One slide clip lies flat just to the right of the bag, on the right arm's side and clear of the food bowl spot, its open end toward the right arm. A sprung clip that has to be squeezed open is

not usable: the gripper cannot squeeze it and place it at the same time.

10. One scoop lies flat on the scoop spot, bowl down, handle toward the right arm. It is wide and shallow, so a quarter turn empties it.
11. One pitcher stands upright on its stand, forward on the left beside the mat, handle toward the left arm, filled near the top so the stream starts within a quarter turn of tilt, and light enough for the gripper to carry it level when full. It stands clear of the mat's left edge and of the water bowl spot behind it. Replace a pitcher that needs more than a quarter turn to start a stream.

Workspace layout

Three rows: the scoop at the back, the bag standing between the two bowl spots in the middle, the mat at the front, with the pitcher a step forward on the left. Everything travels forward toward the mat.

Back row:

1. **Scoop spot**, back right, on the right arm's side. Nothing stands at the back left.

Middle row, left to right:

2. **Water bowl spot**, left of the midline.
3. **Bag spot**, center on the midline, between the two bowl spots, the bag standing open with the slide clip flat on bare table just to its right, on the right arm's side. Shared: the right gripper works it, the left gripper holds it.
4. **Food bowl spot**, right of the midline. Both bowl spots stand empty from Step 2 to the end of the episode.

A clear gap of at least a gripper's width sits between the bag and each bowl spot, so a bowl lifts straight up and travels forward to the mat without touching the bag. The clip lies inside the right hand gap, flat on the table and clear of both the bag and the food bowl spot.

Front row:

5. **Pitcher stand**, on the left arm's side, a step forward of the water bowl spot and level with the mat's back edge, clear of the mat's left edge and never on the mat.
6. **Feeding mat**, front center, straddling the midline. The water bowl goes on its left half and the food bowl on its right half.
7. **Crumb spot**, the strip of bare table between the mat's near edge and the front table edge. Empty at the start, one pile of debris at the end.
8. **Cloth spot**, front right beside the mat.

No carry passes over the other bowl or over the bag, and no gripper travels to the far side of the table.

Vocabulary

- **Arm assignment:** the right gripper wipes, scoops, clips and carries the food bowl. The left gripper holds the bag, holds the fold, and carries the water bowl and the pitcher. Both grippers straighten and fold the bag's top edge together in Step 4, each on its own end. Neither reaches onto the other's side to take or carry anything.
- **Hold:** a closed empty gripper pressed onto an object to keep it still. A hold never lifts, carries or moves what it holds.
- **Tool home:** the spot a tool returns to and lies flat on before the next step starts: the cloth spot, the scoop spot, the pitcher stand. A tool is never left in a bowl, in the bag, on the mat or in a gripper between steps.
- **Release point:** the gripper opens over the mat when a bowl is being staged, or over a tool home. Never over the other bowl, over the bag or over open tabletop.
- **Stable placement:** the object stands still after release, flat on its surface, and does not rock, tip, slide or lean on anything.
- **Spill:** kibble or water outside a bowl. It is left where it falls and cleared in the reset, never on camera.
- **Feeding mat:** the flat mat the finished bowls stand on. It is the output of the episode.
- **Debris:** the scattered remains of what has already been broken up, here the loose kibble and dry crumbs left on the mat at episode start. It is spread out, not clumped, and no piece of it is whole or in its proper place. Anything on the mat that is not the mat is debris.
- **Clean mat:** no debris visible anywhere on the mat surface.
- **Crumb pile:** the debris after the wipe, gathered in one pile at the crumb spot.
- **Cloth:** the folded dry cloth used on the mat. It is held so a broad flat side meets the mat, never by a corner and never balled up. It touches the mat and nothing else, ever.
- **Stroke:** one continuous draw of the cloth from the mat's far edge to its near edge, in contact the whole way, carrying debris off the near edge onto the crumb spot.
- **Kibble bag:** the standing bag the kibble is scooped from and that is sealed in Step 4.
- **Straightened edge:** the bag's mouth pinched shut by both grippers, front wall against back wall, so the top edge runs straight and flat across the bag before it is folded.
- **Fold:** the straightened top edge turned down once by both grippers against the front of the bag along its crease and pressed flat. One fold, not a roll.
- **Slide clip:** the sleeve clip that slides over the fold. Its jaws stay open on their own.
- **Sealed bag:** the fold pressed flat, the clip slid across it with the folded edge fully inside it, and no gap into the bag visible.
- **Scoop:** the handled scoop that moves kibble from the bag to the food bowl.
- **Level scoop:** a scoopful with no kibble standing above the scoop rim. A heaped scoop is not level.

- **Food bowl filled:** kibble lying roughly halfway up the inside of the food bowl, well below the rim.
- **Pitcher:** the handled jug the water is poured from.
- **Water bowl filled:** the water line sitting roughly halfway up the inside wall of the water bowl.
- **Spout gap:** visible daylight between the pitcher spout and the bowl rim, held for the whole pour.
- **Quarter turn:** the most any object is tilted in a gripper. Scooping, tipping and pouring all stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Wipe the mat

Goal: the mat is clean, the debris sits in one pile at the crumb spot, and the cloth is back on its spot.

The mat grips the table on its own, so nothing pins it while the wipe runs.

1.1 Take the cloth

- Approach the folded cloth on its spot from above and close on it so a broad flat side hangs below the gripper.

Check: the cloth presents a flat face and does not hang by a corner or slip. If it does, lay it back on its spot and take it again. Make no more than two corrective takes. If it still will not sit flat, end the episode and correct the cloth during reset.

1.2 Wipe far to near, stepping left to right

- Lay the cloth flat on the mat at the far edge, on the mat's left side.
- Draw it straight toward you to the near edge, flat on the surface the whole way, and carry the debris off the near edge onto the crumb spot.
- Lift the cloth clear at the end of the stroke and carry it back to the far edge through the air. Never draw it back across the mat.
- Step each stroke to the right of the last, overlapping it, and work across until the whole surface has been covered.
- Push debris off the near edge only. Never toward the bowl spots, never off the left or right edge, and never into a pile on the mat.

Debris is moved by the cloth and by nothing else. A gripper never touches it.

Check: no debris is visible anywhere on the mat and all of it sits in one pile at the crumb spot. If any remains, make one more stroke over that part of the mat. If the mat has slid, bunched or lifted, end the episode and fix the mat during reset.

1.3 Park the cloth

- Carry the cloth back to the cloth spot, lay it flat, release, and lift clear.

Check: the cloth lies fully on its spot, not draped over the mat, a bowl or the table edge.

Step 2: Stage the two empty bowls

Goal: both bowls stand empty on the clean mat, water bowl on the left half, food bowl on the right half, with a clear gap between them.

One bowl per carry, water bowl first. Each bowl is carried by the arm on its own side and travels straight forward onto its own half of the mat, outside the standing bag between the spots.

Neither passes over the other, and neither passes over the bag.

- With the **left gripper**, close on the water bowl's rim, square to the bowl so it hangs upright and level. The rim is the grasp: the grippers cannot span a bowl wall securely.
- Lift it straight up clear of the table, carry it forward level without tilting or swinging it, clear of the bag on its right, lower it onto the left half of the mat, and release once it stands on its own.
- With the **right gripper**, take the food bowl the same way and set it on the right half of the mat.
- Leave a clear gap between the bowls. They do not touch, neither overhangs a mat edge, and neither stands on the crumb pile.

Check: both bowls stand empty and upright, fully on the mat, with a gap between them, and both are stable. If a bowl overhangs an edge, touches the other, or rocks, the arm that placed it lifts it clear and places it again. Make no more than one correction per bowl.

Step 3: Scoop the kibble into the food bowl

Goal: two level scoops of kibble in the food bowl, the bag still standing open, and the scoop back on its spot.

3.1 Hold the bag and take the scoop

- With the **left gripper** closed and empty, come down on the left side of the bag's top edge and press it against the table. Take this hold **before the scoop goes into the bag** and keep it until the second scoop has been tipped.
- With the **right gripper**, close on the scoop handle end, lift it straight up off the scoop spot, and carry it to the bag.

The bag is light and the scoop drags on the way out, so an unheld bag lifts with the scoop or tips over. The bag stands on the midline so each arm meets it from its own side: the left gripper on its left, the right gripper working down into it from its right.

3.2 Scoop and tip, twice

- Lower the scoop into the kibble bowl down, and push it forward through the kibble until it is full.
- Draw the scoop up the inside wall of the bag as you lift, so any heap falls back in, until no kibble stands above the rim.
- Lift the scoop clear of the bag, keep it level, and carry it forward and right to the food bowl.
- Bring the scoop over the food bowl mouth, inside the ring of the rim and low to it. Stop, then tilt it a quarter turn until the kibble runs out.
- Bring the scoop back to level before it leaves the bowl mouth.
- Repeat once, for **two scoops in total**. Nothing else changes between them.

The scoop never passes over the water bowl, loaded or empty.

3.3 Park the scoop

- With the **right gripper**, carry the scoop back to the scoop spot, lay it flat bowl down, and release.
- The **left gripper** releases the bag and lifts clear only now.

Check: kibble lies roughly halfway up the inside of the food bowl, the bag still stands open, and the scoop lies flat on its spot. A short fill is not corrected with a third scoop: leave it and let review flag the episode.

Step 4: Seal the bag

Goal: the bag's mouth straightened, folded down once, clipped shut, and still standing at the bag spot.

Both arms work the bag from the front, each on its own side of it. The bag stays on the table throughout and is never lifted.

4.1 Straighten the top edge

- Bring both grippers down on the bag's top edge, the **left gripper** on its left end and the **right gripper** on its right end, and pinch the front and back walls together on each side.
- Draw the two holds toward each other along the edge until the mouth is closed and the top edge runs straight and flat across the bag, front wall against back wall along the crease.

The bag stands open at this point and its mouth is too wide to turn down cleanly with one gripper: a single arm folds one corner and leaves the other side gaping. Both arms flatten it first, so the fold has one straight edge to land on.

4.2 Fold the edge down

- With both grippers still holding their own end of the straightened edge, turn it down together once against the front of the bag along the crease, keeping the edge level the whole way.

- Press the fold flat with both grippers, each on its own end.

Both ends move as one. A fold turned down by one arm while the other end is left standing is not a fold.

4.3 Clip the fold

- The **left gripper** stays closed on the left end of the fold and holds it shut. This hold does not move again until the clip is on.
- With the **right gripper**, release the fold, take the slide clip from the right of the bag, and bring it to the right end of the fold.
- Slide the clip along the fold toward the middle until the folded edge sits fully inside it, then release and lift clear.
- The **left gripper** releases the fold only once the clip is on and holding.

Check: the clip sits across the fold with the folded edge fully inside it, no gap into the bag is visible, and the bag still stands. If the clip sits on a corner of the fold or has come away, take it off with the right gripper, lay it back to the right of the bag, hold the fold again, and clip it once more. Make no more than one correction.

Step 5: Pour the water

Goal: the water bowl filled to roughly half, the pitcher back on its stand, and nothing spilled.

The water bowl is not held while the pour runs: the stream goes into the middle of the bowl, which does not push it.

5.1 Lift and carry the pitcher

- With the **left gripper**, close on the middle of the pitcher's handle span and lift it straight up off its stand before any horizontal motion.
- Carry it forward and right to the water bowl, upright the whole way, until the spout sits over the bowl mouth at the spout gap. The carry is short: the stand sits beside the mat.

The pitcher never passes over the food bowl and never over the bag.

5.2 Pour

- Tilt the pitcher forward, no more than a quarter turn, and start the stream into the middle of the bowl.
- Hold the tilt until the water line reaches roughly halfway up the inside wall, then bring the pitcher back upright to stop the stream.
- Hold the spout over the bowl for 2 seconds so the last drips fall inside.

One pour, and one only. An underfill is not topped up and an overfill is not poured off. Either one is tagged and the episode continues.

5.3 Park the pitcher

- Carry the upright pitcher back over its stand, lower it until the base is seated, then open the gripper and lift clear.

Check: the water sits roughly halfway up the inside wall of the bowl, no water is on the mat, on the table, or in the food bowl, and the pitcher stands flat on its stand without rocking. If it rocks, lift it clear and seat it again, once.

Step 6: End the episode

- Confirm the mat: no debris on it, the debris in one pile at the crumb spot.
- Confirm the bowls: both fully on the mat, water on the left half, food on the right half, each filled to roughly half, with a clear gap between them.
- Confirm the bag: folded, clipped and standing.
- Confirm the tools: the cloth flat on its spot, the scoop flat on its spot, the pitcher upright on its stand.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return both arms home with their grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Tip the kibble from the food bowl back into the bag, empty the water bowl, dry both, and stand them back on their spots either side of the bag: water bowl left of the midline, food bowl right of it, each a clear gap from the bag.
2. Slide the clip off the bag, open the top edge so the bag stands open again, crease it once by hand, and lay the clip flat to the right of the bag with its open end toward the right arm.
3. Top up the bag so it holds well more than two scoops, and refill the pitcher near the top.
4. Sweep the crumb pile off the crumb spot into the bin, then scatter fresh debris across the mat, spread out rather than clumped and clear of the mat edges.
5. Wipe any spilled kibble or water off the tabletop and off the mat, and dry both.
6. Lay the scoop flat on its spot bowl down, stand the pitcher on its stand forward on the left with the handle toward the left arm, and lay a folded dry cloth flat on the cloth spot. Replace a cloth that is damp or loaded with dust.
7. Confirm the mat lies flat and still grips the table, neither bowl is cracked or warped, the clip still holds a fold, and the pitcher does not drip.
8. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a bowl goes onto the mat before the wipe is finished or while debris is still on the mat surface, the **right gripper** carries the food bowl onto the mat before the **left gripper** has placed the water bowl, both bowls are moved in one carry, the right gripper puts the scoop into the bag before both bowls are staged, the clip goes on before the second scoop is tipped, the left gripper starts the pour before the bag is sealed, or a step is skipped.
- **SOP rule broken:** Steps 1 to 5, wipe, stage, scoop, seal, pour, in that order, one bowl per carry with the water bowl first.
- **Coaching note:** wipe it, stage it, scoop it, seal it, pour it. Water bowl leads, one bowl per trip, and nothing goes onto a mat that is not clean.

Violation: Arms swapped roles

- **Visible cue:** the **left gripper** takes or carries the cloth, the scoop, the clip, or the food bowl, or does the wipe or the clip; the **right gripper** takes or carries the pitcher or the water bowl; or either gripper reaches onto the other arm's side for an object that belongs to that arm. The two armed straighten and fold in Step 4 is not a swap: there each gripper works its own end of the bag.
- **SOP rule broken:** Steps 1 to 5, the right gripper wipes, scoops, clips, and carries the food bowl, and the left gripper holds the bag, holds the fold, and carries the water bowl and the pitcher, with both arms folding the bag together in Step 4.
- **Coaching note:** if an arm is reaching across, the wrong arm has been chosen. Right works and carries the food bowl, left holds and owns the water side.

Violation: Cloth held wrong, or the mat wiped without it

- **Visible cue:** the **right gripper** holds the cloth by a corner or balled up so only a narrow line of it meets the mat, or wipes the mat with the bare gripper.

- **SOP rule broken:** Step 1.1, the right gripper holds the cloth so a broad flat side meets the mat.
- **Coaching note:** a broad flat face on the mat. A bare gripper smears the debris instead of carrying it.

Violation: Wipe stroke wrong

- **Visible cue:** the cloth lifts part way down a stroke, a stroke stops short of the near edge, strokes do not overlap, parts of the mat are never touched by the cloth, a stroke pushes debris toward the bowl spots or off a side edge or into a pile on the mat, the strokes step right to left, or the **right gripper** draws the cloth back to the far edge still in contact with the mat.
- **SOP rule broken:** Step 1.2, overlapping strokes in contact from the far edge to the near edge, stepping left to right, debris off the near edge only, and the cloth lifted clear and carried back through the air.
- **Coaching note:** every stroke pulls toward the front edge, the strokes walk left to right, and the return goes through the air. The back stroke is the most common reason a mat never gets clean.

Violation: Debris handled by a gripper

- **Visible cue:** the **left gripper** or the **right gripper** picks up, pinches, brushes, or pushes debris, on the mat or at the crumb spot.
- **SOP rule broken:** Step 1.2, debris is moved by the cloth and by nothing else.
- **Coaching note:** the cloth does this job. A gripper never touches debris.

Violation: Bowl placed wrong on the mat

- **Visible cue:** the water bowl ends on the right half or the food bowl on the left half, part of a bowl hangs off a mat edge, the bowls touch, or a bowl stands on the crumb pile.
- **SOP rule broken:** Step 2, each bowl stands fully on its own half of the mat with a clear gap between them.
- **Coaching note:** look at the placement before the gripper withdraws.

Violation: Bowl released before it is stable

- **Visible cue:** a gripper opens while the bowl is still moving, or the bowl rocks, tips, or slides after release.
- **SOP rule broken:** Step 2, the bowl stands still after release, flat on the mat, and does not rock.
- **Coaching note:** let the bowl take its own weight before the gripper opens.

Violation: Bag not held, lifted, or knocked over

- **Visible cue:** the **left gripper** is off the bag at any point between the scoop entering the bag and the second scoop being tipped; the bag lifts, shifts, or tips as the scoop is drawn out; or the bag leaves the table, falls on its side, or is knocked out of the bag spot.
- **SOP rule broken:** Steps 3.1 and 4, the left gripper holds the left side of the bag's top edge until the second scoop has been tipped, and the bag stays standing on the table throughout.
- **Coaching note:** the holding arm holds through both scoops, not just the first. The bag is worked in place, and nothing in this task requires picking it up.

Violation: Bag folded or clipped wrong

- **Visible cue:** the fold is turned down while the bag's mouth is still open, or one gripper folds its end alone while the other end stands open; the top edge is folded more than once or rolled; the clip comes on with no fold, runs across the open mouth, ends on a corner of the fold, or is brought from the left end; a gap into the bag is still visible after the clip is on; or the **left gripper** comes off the fold while the clip is still being positioned.
- **SOP rule broken:** Step 4, both grippers pinch the mouth shut and straighten the top edge, then turn it down together once along its crease and press it flat, the clip slides on from the right end with the folded edge fully inside it, and the left gripper releases the fold only once the clip is holding.
- **Coaching note:** straighten it with both, fold it with both, then only the right arm leaves for the clip. One fold, pressed flat, clip on from the right end. The holding arm releases last, after the clip, every time.

Violation: Scoop count wrong

- **Visible cue:** fewer or more than two scoops are tipped into the food bowl, or the **right gripper** takes a third scoop to correct a short fill.
- **SOP rule broken:** Step 3.2, two scoops in total.
- **Coaching note:** the count is fixed at two. A correction scoop makes the episode unlearnable.

Violation: Scoop heaped, or tipped from the wrong place

- **Visible cue:** kibble stands above the scoop rim as the **right gripper** lifts the scoop out of the bag, with no draw up the inside wall to level it; the scoop tilts before it is over the bowl mouth, is held outside the ring of the rim, or is held high so the kibble falls a visible distance; or the scoop leaves the bowl mouth before it is back to level.
- **SOP rule broken:** Step 3.2, level the scoop against the inside wall of the bag, bring it inside the ring of the rim and low to it before tilting, and bring it back to level before it leaves the bowl mouth.
- **Coaching note:** ride the inside wall on the way up. Finish the carry, stop, then tilt. Level first, then leave.

Violation: Pour done wrong

- **Visible cue:** the **left gripper** tips the pitcher during the carry, water leaves the spout before the spout is over the bowl mouth, or the spout rests on the bowl rim or sits inside the bowl with no daylight between them.
- **SOP rule broken:** Steps 5.1 and 5.2, carry the pitcher upright, set the spout over the bowl at the spout gap, then tilt.
- **Coaching note:** finish the carry, set the gap, then tilt. The gap is set before the stream starts, not during it.

Violation: Fill level out of range

- **Visible cue:** water spills over the water bowl rim, the pour stops with the line visibly short of halfway up the inside wall, or the food bowl ends visibly under or over half.
- **SOP rule broken:** Steps 3.2 and 5.2, kibble roughly halfway up the inside of the food bowl and the water line roughly halfway up the inside wall of the water bowl.
- **Coaching note:** the eyes stay on the rising water line rather than on the pitcher.

Violation: Second pour or pour off

- **Visible cue:** the **left gripper** pours into the water bowl a second time after the pitcher has come back upright, or tips water back out of the bowl to correct an overfill.
- **SOP rule broken:** Step 5.2, one pour, and one only.
- **Coaching note:** the fill is one continuous pour. A correction pour makes the episode unlearnable.

Violation: Spill cleaned up

- **Visible cue:** after kibble or water lands outside a bowl, a gripper or the cloth is used to sweep, wipe, push, or pick it up.
- **SOP rule broken:** Steps 3 and 5, a spill is left where it falls and is cleared in the reset.
- **Coaching note:** cleanup motions are not part of the task. The spill is data, the cleanup is noise.

Violation: A tool carried over the other bowl

- **Visible cue:** the scoop passes over the water bowl, the pitcher passes over the food bowl or over the standing bag, or an empty bowl is carried over the bag on its way to the mat.
- **SOP rule broken:** Steps 2, 3.2 and 5.1, the scoop never passes over the water bowl, the pitcher never over the food bowl, and nothing is carried over the bag.
- **Coaching note:** each tool stays over its own bowl, and the bag between the spots is gone around, never over. Route around, not across.

Violation: Bad pick, carry, or release

- **Visible cue:** an object moves from one gripper to the other without being set down first, both grippers hold an object at once, a gripper shifts or seats its hold again while the object is off the table, a bowl or the pitcher or the scoop or the clip slides along the table with no daylight under it, a bowl hangs tilted in the gripper because the rim grasp was taken off square, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 5, one object at a time, lifted straight up clear of the surface before any horizontal motion, a bowl gripped square on its rim, tilted no more than a quarter turn, and never passed between grippers.
- **Coaching note:** set it down to change the grip, and lift before the carry. A bowl that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a bowl, the clip, the mat, or a tool is shoved, nudged, or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 5, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** a gripper opens while the pitcher base is still above the stand; the scoop is left in the bag or in a bowl, the cloth is left on the mat or the tabletop, or the pitcher is set down off its stand; or a tool is still in a gripper when the next step begins.
- **SOP rule broken:** Steps 1.3, 3.3, and 5.3, each tool returns to its home and lies flat there before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower until it touches, then release.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, a gripper or a carried object strikes the bag, a bowl, the pitcher, the mat, or the crumb pile and knocks, tips, slides, or displaces it; the two arms strike each other over the bag; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 5, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. On the bag, the working arm moves and the holding arm stays put.

Violation: Manipulation outside the camera frame

- **Visible cue:** a stroke, a placement, a scoop, a clip, or a pour happens partly or wholly outside the environment camera frame.
- **SOP rule broken:** Steps 1 to 5, every manipulation happens inside the camera frame.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a bowl still on its bowl spot, a bowl unfilled, the bag unsealed, a tool out of its home, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 6, confirm the mat, the bowls, the bag, and the tools, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **A spot sits outside an arm's comfortable reach,** so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- **Defective mat:** one that slides, bunches or lifts under a correct stroke, or that will not lie flat. Weight or replace it before the next episode.
- **Defective bowl:** one that will not sit flat, whose rim the gripper cannot close on squarely, or that is cracked or warped.
- **Defective clip:** one whose jaws will not stay open on their own, or that will not hold a fold.
- **Defective bag:** one that will not stand open on its own, whose top edge will not hold a crease, or that is too tall for the scoop to reach the kibble from above.
- **Defective pitcher:** one that drips from the spout, that needs more than a quarter turn to start a stream, or that is too heavy for the gripper to carry level when full.
- **Damp or dusty cloth at episode start.**
- **Bag holding too little kibble for two level scoops,** or a pitcher holding too little water for a half full bowl. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take the cloth from its spot
2. Wipe one stroke down the mat

3. Return the cloth to its spot
4. Move one empty bowl onto the mat
5. Take the scoop from its spot
6. Hold the bag steady
7. Scoop one level scoop of kibble from the bag
8. Tip one scoop of kibble into the food bowl
9. Return the scoop to its spot
10. Straighten the top of the kibble bag with both grippers
11. Fold the top of the kibble bag down with both grippers
12. Slide the clip onto the fold
13. Lift the pitcher from its stand
14. Pour water into the water bowl
15. Return the pitcher to its stand
16. Return both arms home and end the episode